Instrumentation Laboratory
A Werfen Company

OPERATIONAL QUALIFICATION
IL ACL TOP® FAMILY
SYSTEMS

Prepared by:
Instrumentation Laboratory

Prepared for:
Facility name: DUSHANBE INTERNATIONAL HOSPITAL
Address:

Equipment location: LABORATORY
Date: 10/10/26

Equipment installed

Model number: ACL TOP 350 CTS
Serial number: 26025983
Date of installation: 10/10/26

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1.0 OQ Approval

The Operational Qualification is a controlled document in the Document Control System and has been prepared and approved by Instrumentation Laboratory Company (IL).

Instrumentation Laboratory
180 Hartwell Road
Bedford, MA. 01730

2.0 OQ Execution

The Operational Qualification has been conducted and completed by Instrumentation Laboratory Company or an authorized dealer.

Name / Position	Company	Date
ILYA NOVOPASHIN ENGINEER	LABTECH DIAGNOSTICS	10/10/2026

3.0 OQ Documentation

A copy of the Operational Qualification documentation will be provided to:

Facility: NUSAIBE INTERNATIONAL HOSPITAL
Name: PATHO BORVAN
Position: HEAD OF LABORATORY

4.0 OQ Scope

The scope of the Operational Qualification defines the requirements and criteria of acceptance which verifies the operation and functionality of the equipment. Equipment analytical performance verification will include successful calibration and analysis of control material for a limited group of assays. The steps outlined in this procedure shall be followed as written with results recorded. Any abnormalities along with corrective actions will be noted in the Operational Qualification documentation. Successful completion of this protocol ensures that this instrument is performing to Manufacturer's specifications.

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5.0 Operational Qualification Requirements

5.1 Verify that Installation Qualification has been successfully completed.

Facility/Company	Title / Position	Name (print)	Signature
LABTECH DIAGNOSTICS	ENGINEER	ILYA NOVOPASHIN	

Notes: _____

5.2 Perform the following operational checks;

a) Occlusion Test

UI	Replicate	Result (pass/fail)	Corrective Actions Applied
8ul	1		
8ul	2		
50ul	1		
50ul	2		

Notes: _____

b) Rinse Flow Check – Sample, R1, R2

Rinse Flow Check	Specification	Result	Corrective Actions Applied
Sample	4.54 ml to 6.61 ml		
R1	4.54 ml to 6.61 ml		
R2 (if applicable)	4.54 ml to 6.61 ml		

Notes: _____

c) Arm Coordinate adjustment - Sample, R1, R2

Arm coordinate adjustment	Result (pass/fail)	Corrective Actions Applied
Sample	PASS	
R1	PASS	
R2 (if applicable)		

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Notes: _____

d) _____ Module Temperature.

Verify temperatures displayed are within specified range.

Module	Specified Temperature Range	Displayed Temperature	Corrective Actions Applied
ORU #1	36.8°C – 37.2°C	37.0	
ORU #2	36.8°C – 37.2°C	37,0	
ORU #3 (if applicable)	36.8°C – 37.2°C	–	
ORU #4 (if applicable)	36.8°C – 37.2°C	–	
Incubator #1	36.7°C – 37.3°C	37	
Incubator #2	36.7°C – 37.3°C	–	
Reagent Module	14.5°C – 15.5°C	15,12	
Shuttle	36.5°C – 37.5°C	37,0	
R1 Probe	36.0°C – 38.5°C	36,95	
R2 Probe (if applicable)	36.0°C – 38.5°C		

Notes: _____

e) ORU Air Blank (all ORUs)

ORU Air Blank	Result (pass/fail)	Corrective Actions Applied
ORU #1	PASS	
ORU #2	PASS	
ORU #3 (if applicable)		
ORU #4 (if applicable)		

Notes: _____

f) Factor Diluent Blank (all ORUs)

Factor Diluent Blank	Result (pass/fail)	Corrective Actions Applied
ORU #1	PASS	
ORU #2	PASS	
ORU #3 (if applicable)		
ORU #4 (if applicable)		

Notes: _____

5.3 Performance Checks

a) Fluidic Precision Check. Perform fluidic precision check for each arm.

Arm	Mean absorbance 100% dilution 966.00 to 1100.00	Slope 9.6 to 11.0	Intercept -8.00 to +8.00	R^2 ≥0.9989	Corrective Actions Applied
Sample					
R1					
R2 (if applicable)					

Notes: _____

5.4 Calibration and Controls

a) Perform calibration for AT and if applicable PT. Run normal and low abnormal control material for AT. Run normal and abnormal high and low control material for PT, and APTT assays.
b) If calibration fails, repeat calibration up to 3 times.
c) If calibration fails more than 3 times, complete necessary corrective actions.

Calibrators and Reagents: PT, AT, APTT

Material	Lot Number	Expiration Date	Part Number
Factor Diluent	N0853783	2028-02-29	00097576000
Calibration Plasma	L0653531	2028-06	00200003710
PT Reagent	N1158589	2027-12	00200003051
AT Liq FXa			
AT Liq Substrate			
APTT Reagent	N1057965	2027-10	00200006800
APTT CaCL2	N1057966	2028-10	00200006800

Calibration Data:

PT Calibration	Value 1 (sec)	Value 2 (sec)	Value 3 (sec)
PT 100%	11,6	11,5	11,6
PT 50%	17,3	17,1	17,5
PT 25%	30,8	30,8	30,5

PT Calibration R^2 Specification	PT Calibration R^2 Result	Corrective Actions Necessary?
≥ 0.980	0,998	NO

Notes: _____

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AT Calibration	Value 1 (mA/min)	Value 2 (mA/min)	Value 3 (mA/min)
AT _____% (*)			
AT _____% (*)			
AT _____% (*)			

AT Calibration R^2 Specification	AT Calibration R^2 Result	Corrective Actions Necessary?
≥ 0.980		

(*) Indicate target value of calibrator used for calibration of 100%, 50% and 25%.

Notes: _____

Control Material:

Control Material	Lot Number	Expiration Date	Part Number
PT Normal Control	N0160239	01-2029	00200031/00
PT Abnormal Low	N1258760	12-2028	00200032/00
PT Abnormal High	N1158384	11-2028	00200035/00
AT Normal Control			
AT Abnormal Low			
APTT Normal Control	N0160235	01-2029	00200031/00
APTT Abnormal Low	N1258760	12-2028	00200032/00
APTT Abnormal High	N1158384	11-2028	00200035/00

Control Results:

Normal Control

PT	Control Range	Control Result	Corrective Actions Necessary?
Seconds	4 sec	11,5	NO
AT			
% Activity			
APTT	2 sec	28,4	NO
Seconds			

Notes: _____

Abnormal Control Low

PT	Control Range	Control Result	Corrective Actions Necessary?
Seconds	2,8	2,2	NO
AT			
% Activity			
APTT			
Seconds	13,2	44,3	No

Notes: _____

Abnormal Control High

PT	Control Range	Control Result	Corrective Actions Necessary?
Seconds	14,8	32,7	NO
APTT			
Seconds	16	52,8	NO

Notes: _____

6.0 Training

Verify that Key Operator has received necessary instruction/training for operation and routine user maintenance based on site test menu.

Operator Name (print)	Title / Position	Signature
Hristova Vyara	deputy of HOD Lab	

Notes: _____

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7.0 Operational Supplies

The use of non-IL brand reagents or supplies for testing may cause a clinically significant degradation of performance and results. IL does not assume any obligation or warranty engagement concerning precision and/or accuracy of the measurements nor for any damage to the instrument directly or indirectly resulting from the use of reagents, consumables and expendable supplies different from those produced by IL.

The facility acknowledges that instrument performance is only validated using reagents, consumables and expendable supplies produced by IL.

Facility	Name (print)	Title/Position	Signature
✓			

8.0 OQ Acceptance

The Operational Qualification (OQ) has been successfully completed by Instrumentation Laboratory Company or authorized dealer by;

Company	Name (print)	Title / Position	Signature	Date
LABTECH DIAGNOSTICS	NOVOPASHIN	ENGINEER		10/10/26

and the facility indicated below accepts installation of the equipment.

Facility	Name (print)	Title / Position	Signature	Date
✓				

Notes: _____